

Scattering of Sound by a Sphere

Abstract

The problem of scattering of acoustic plane waves by a rigid sphere is considered. The quantities computed include S and Q , the intensity of the scattered wave in different directions, and the total energy scattered. Graphs of both quantities for different ratios ka between the radius and wavelength have been presented and discussed.

1 Introduction

We are concerned the propagation of an acoustic wave in a collision course, and quantities concerning the scattered waves which makes this a direct scattering problem. Consider a plane sound wave of wavelength λ (and wavevector $k = 2\pi/\lambda$) which meets a rigid solid sphere of radius a at the origin. This situation has a simple geometry, and the understanding required in being able to solve such a simple system is essential for solving more complex systems, as we can model the constituents of the complex system by manipulating simple systems. As the sphere is rigid, it imposes specific boundary conditions (Neumann boundary conditions) on the solutions to the three-dimensional wave equation. The physical interpretation of these boundary conditions manifest by specifying the velocity at the surface of the sphere. To understand the analytical solution requires mathematical prowess in the study of partial differential equations. The complicated expressions which are derived as solutions, do not elucidate the situation. Thus we take a numerical approach by evaluating these expressions in order to understand the behaviour of the scattered wave.

The upcoming section serves as an introduction to the computed values, which is followed by the method undertaken for numerical evaluation. Then we analyse the gathered results, presented with plots, which leads onto a discussion about the method, including the conclusion.

2 The Scattered Wave

The quantities that are interesting are; the ratio S between the intensity of the scattered wave and incident wave in different directions θ at a large distance r from the sphere; the ratio Q between the total energy scattered in all directions and the incident intensity. These can be shown to be given by the expressions:

$$S = \frac{1}{k^2 r^2} \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} (2m+1)(2n+1) \cos(\delta'_m - \delta'_n) \sin(\delta'_m) \sin(\delta'_n) P_m \cos(\theta) P_n \cos(\theta)$$

$$Q = 4\pi a^2 \sum_{n=0}^{\infty} \frac{2n+1}{(ka)^2} \sin^2[\delta'_n(ka)]$$

where the numbers δ'_x are defined by

$$\tan(\delta'_x) = -\frac{j'_n(x)}{y'_n(x)}$$

in which $j'_n(x)$, $y'_n(x)$ are the derivatives of the spherical Bessel functions of the second kind. The presence of the derivatives are attributed to the Neumann boundary conditions on the surface of the sphere. Note that ka is proportional to the ratio of the radius of the sphere a and the wavelength λ , with the proportionality constant being 2π . It is dimensionless as required for the argument of trigonometric functions. Therefore, without loss of generality, we can take the radius to be of unit length i.e $a = 1$

To bring some familiarity to these expressions, recall that a plane wave with wave-vector $\vec{k}$ travelling in a directional parallel to $\vec{r}$ has a functional form of $\psi \propto e^{i\vec{k} \cdot \vec{r}}$

Via the property of Bessel functions, and considering the source of the plane waves to be at an infinite distance from the plane wave under consideration, one can obtain the expansion of the plane wave as a sum of spherical waves

$$e^{ikr \cos(\theta)} = \sum_{n=0}^{\infty} (2n+1) i^n P_n(\cos \theta) j_n(kr)$$

where j_n are the spherical Bessel functions of the second kind, and P_n are the Legendre polynomials. This is for the case where the plane wave vector is parallel to the line connecting the source to the plane wave. Thus the angle θ is the spherical polar angle of r . In the general case, θ should be replaced by $\hat{k} \cdot \hat{r}$

3 Numerical Evaluation

3.1 Algorithms for the functions

The procedure in evaluating the given expressions for Q and S involved writing scripts to evaluate the Legendre polynomial and the spherical Bessel functions. All the functions were evaluated using recurrence relations.

The Legendre Polynomial were evaluated using Bonnet's recursion formula:

$$nP_n(x) = (2n-1)xP_{n-1}(x) - (n-1)P_{n-2}(x)$$

with the initial values $P_0 = 1$ and $P_1 = x$. Thus, for a given x , we know the values of P_0 and P_1 . With the recurrence relation being stable, the upwards recursion could be used to evaluate P_n .

The spherical Bessel functions $y_n(x)$ followed a similar process with the initial values:

$$y_0 = -\frac{\cos(x)}{x} \quad y_1 = -\frac{\cos(x)}{x^2} - \frac{\sin(x)}{x}$$

and the stable recursion relation:

$$y_n(x) = \frac{2n-1}{x}y_{n-1}(x) - y_{n-2}(x)$$

The spherical Bessel function j_n posed problems as the upwards recurrence relation is unstable.

Hence, we use downwards recursion. We begin by setting false initial values for high order Bessel functions j_N and j_{N-1} where N is a large enough integer such that the contribution of the true values j_N and j_{N-1} are negligible in our sum. Then we can use downwards recursion to acquire the values for $j_{N-2}, j_{N-3} \dots \dots j_1, j_0$. As we are computing large values from small, there is no problem with accuracy. The recursion relation is of the form:

$$j_n(x) = \frac{2n+3}{x}j_{n+1}(x) - j_{n+2}(x)$$

We know the correct initial values for j_n , which should be

$$j_0 = \frac{\sin(x)}{x}, \quad j_1 = \frac{\sin(x)}{x^2} - \frac{\cos(x)}{x}$$

Thus, we can scale our false values to give the correct values for j_n , by calculating the scaling factor via the correct initial values.

This method serves our purpose as the incorrect values are so small in magnitude and thus insignificant. However, we receive incorrect values when this method of downward recursion is used to calculate Bessel functions of order greater than N , leading to large numbers incorrect by many magnitudes. Thus, this recursive method cannot be used. To calculate Bessel functions j_B of higher order B , N must be greater than B .

For the numbers δ'_x , the derivatives of the Bessel functions $j'_n(x)$, $y'_n(x)$ are necessary, and were acquired using the recursive relation:

$$(2n + 1) j'_n(x) = n j_{n-1}(x) - (n + 1) j_{n+1}(x)$$

The relation for y'_n is similar.

Creating the compound expression for Q and S, resulted in long computation time with numerical errors which we were unable to resolve, thus functions from scientific packages have been used. Specifically, the python package scipy. However, the verification (and comment of the errors and limitations) of the functions created using the methods above has been documented with the code in the appendix.

4 Q - the total energy scattered

The expression for Q can be re-written as

$$Q = \frac{\lambda^2}{\pi} \sum_{n=0}^{\infty} (2n + 1) \sin^2[\delta'_n(ka)]$$

where the factor of $(ka)^2$ has been taken out of the summand and evaluated with the pre-factor. Thus Q 's explicit functional dependence on a vanishes, and is only dependent on the ratio ka through the Bessel functions. As a acts only as a multiplicative factor on the wavevector k , by considering $a = 1$ (as stated earlier) is sufficient to observe the behaviour of Q. This can be verified by considering Q as also a function of radius, but is not presented here as it is trivial. Thus we only consider Q as a function of ka , which is the ratio between the radius of the sphere and wavelength.

A simple while loop (in Python) was used to evaluate the summation expression for Q , for a given value of ka . In the analytical expression the sum continues ad infinitum. Numerically, this is not possible. Thus the summand did not contribute to the sum when its magnitude was insignificant. For our purposes, this is when the resolution of the resultant graph for Q was indistinguishable. We wanted to observe the limiting behaviour of Q as ka increased, and thus we considered Bessel functions (via $\delta'_n(ka)$) of high orders. This lead to numerical errors, visible as steep dips in the plot of Q. To correct for this, we decreased the threshold for the contributing summand. This lead to contributing terms which were of the magnitude of 10^{-15} , but no smaller. A lower threshold could have been used if we did not consider a large ka .

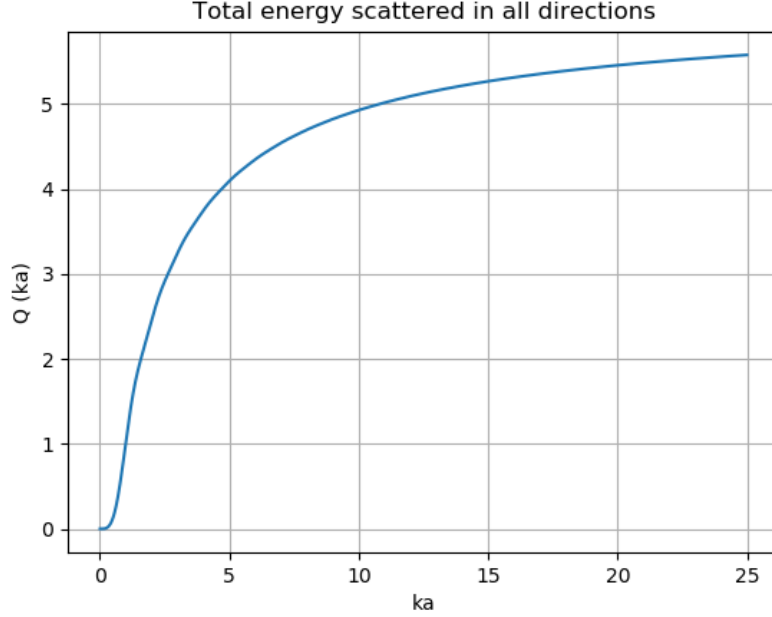


Figure 1: This figure shows the total energy scattered from the sphere Q , as a function of the ratio between the radius of the sphere and the wavelength of the incident wave, ka .

Figure 1 above shows $Q(ka)$, which looks like a monotonically increasing function. From this we can infer that the scattered energy is large if the wavelength is small in comparison to the radius of the sphere. However, as the ratio ka increases, it is easy to see that $Q(ka)$ approaches a limiting value. This must be the case as we require conservation of energy.

The theoretical result for the limiting behaviour of Q as $ka \rightarrow 0$ can be shown to be given by $Q \rightarrow \frac{4}{9}\pi a^2(ka)^4$. We were not able to test to verify this result.

5 S - scattered intensity in different directions

The expression for S contains a factor $\frac{1}{r^2}$ outside of the summand, where r denotes the distance away from the sphere, and thus we know the scattered intensity at a distance from the sphere follows an inverse square law. Thus we can considering the distance to be r to be a unit length away from the sphere without loss of generality. We are interested in the scattered intensity in different directions from the sphere, for different values of ka .

We considered S to be a function of k and the angles within $[0, 2\pi]$, returning a single value which was the scattered intensity. Within the function, the algorithm used to evaluate S computed one sum after another. A private function was used to represent the summand, which took the sum indices m and n as inputs, and returned the value of the summand. Again, as the summation runs ad infinitum, we must limit the contribution to the sum beyond a floor value. Whilst computing expression, specifically the summand, we observed that the magnitude of the summand became very small (of the order 10^{-12}) for indices greater than 10. Thus, we only summed over m and n up to and including 10. This resulted in a reasonable computational time, with smooth plots.

5.1 ka less than 1

We first considered the radius of the sphere to be much less than the wavelength of the wave.

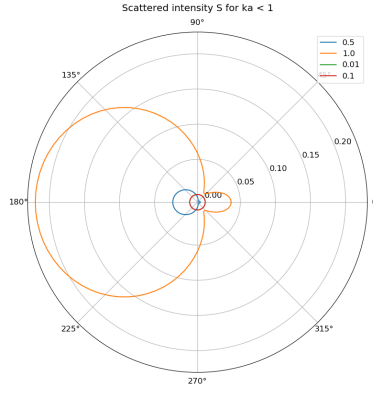


Figure 2: This figure shows the scattered wave intensity S in different directions in the form of a polar plot, for different values of ka less than 1. The radial distance from the origin is S . The values of ka are given in the legend.

From the figure above, we deduce that in both cases the wave is scattered backwards, and the profile of the scattering is shaped similar. However, the intensity of the scattered wave S differ in magnitude by a large amount. When the wavelength of the wave is much greater than the radius of the sphere, the intensity of scattering is higher. The scattered waves behind the sphere destructively interfere quickly.

5.2 ka greater than 1

Then we considered the radius of the sphere to be greater than the wavelength of the wave

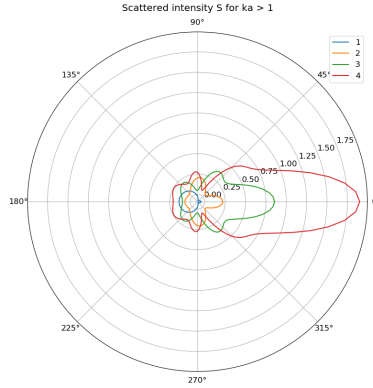


Figure 3: *This figure shows the scattered wave intensity S in different directions in the form of a polar plot, for different values of ka greater than 1. The radial distance from the origin is S . The values of ka are given in the legend.*

The scattered waves are able to reach behind the sphere, with the intensity of the wave increasing as the wavelength decreases. We first considered the radius of the sphere to be much less than the wavelength of the wave. Lobes start to form around the sphere, being symmetrical around the horizontal axis of the sphere. The most prominent region of intensity is right behind the sphere, and can be attributed to the symmetry of the situation. As the wavelength decreases, the surrounding lobes gradually reduces in size and the scattering intensity becomes prominent mostly behind the sphere. The figure below with $ka = 10$ shows this behaviour.

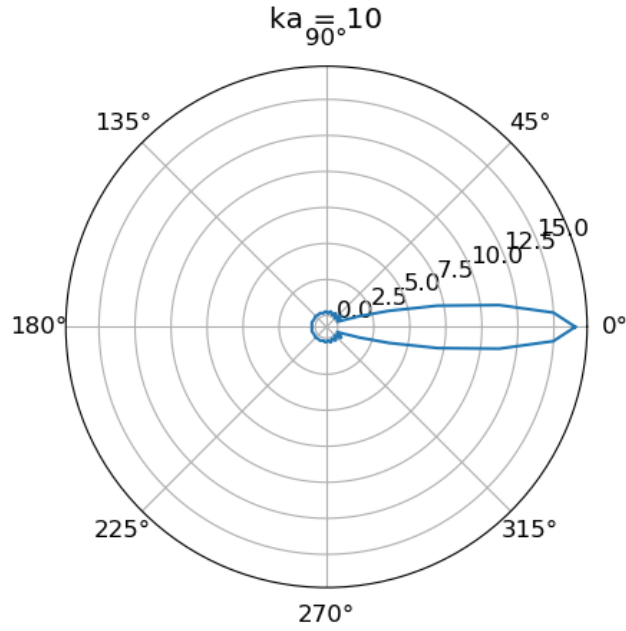


Figure 4: This figure shows the scattered wave intensity S in different directions in the form of a polar plot for $ka = 10$.

6 Conclusions

The problem of scattering of acoustic plane waves by a rigid sphere is considered. The quantities explored include S and Q , the intensity of the scattered wave in different directions, and the total energy scattered. We see that Q approaches a limiting value as ka increases. We also see the difference in behaviour for $ka > 1$ and $ka < 1$.